

Rakesh Purbiya

AI Engineer | Generative AI & Agentic Systems

Surat, Gujarat, India | purbiyarakesh33@gmail.com | +91 8469587067 | github.com/purbiyarakesh33 | linkedin.com/in/rakeshpurbiya-0b7091317

PROFILE

AI Engineer and second-year B.Tech. student at SVNIT building production-oriented AI and machine-learning systems. Focused on Generative AI, agentic workflows, LLM applications, retrieval-augmented generation, and reliable test-driven software automation. Completed the Andrew Ng Machine Learning Specialization and shipped live ML applications.

EDUCATION

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat | B.Tech. in Mechanical Engineering | Aug 2024 - Nov 2028

SELECTED PROJECTS

AI Software Engineering Agent | *Python, LangGraph, Ollama, Docker, Streamlit, GitHub, LangSmith*

Built a multi-agent workflow that investigates software issues, creates plans, reproduces bugs, applies changes in an isolated sandbox, validates tests, and prepares draft pull requests. Added human approval gates, checkpoint-based resume support, GitHub/local repository sources, and Ollama-compatible model backends.

Amazon Electronics Recommender System | *PyTorch, GPU-accelerated matrix factorization, two-tower neural network*

Built collaborative-filtering, content-based, and hybrid recommenders on the Amazon Electronics dataset. Grid search selected pure collaborative filtering as the best model, achieving HitRate@10 of 0.4046. Implemented temporal splits, cold-start filtering, sparse storage, and live Hugging Face deployment.

Credit Card Fraud Detection - Semi-Supervised Learning | *TensorFlow, autoencoder, XGBoost, SMOTE*

Developed a two-stage fraud detector trained on 1.8M+ realistic simulated transactions. The autoencoder learns legitimate behavior and produces anomaly scores that are fused with an XGBoost classifier for the final decision.

Customer Churn Prediction - Telecom | *Python, Scikit-learn, XGBoost, Streamlit*

Compared XGBoost, Random Forest, Logistic Regression, and neural-network classifiers with bias/variance analysis, threshold tuning, and GridSearchCV. Deployed the selected model as a live Streamlit application with 81% accuracy.

TECHNICAL SKILLS

Generative AI: LLM applications, agentic AI, prompt engineering, LangGraph, LangChain, RAG, Self-RAG, Corrective RAG (CRAG), Agentic RAG, embeddings, vector search, reranking, Ollama, LangSmith

Engineering: Python, FastAPI, Streamlit, Docker, Git, GitHub, REST APIs, automated testing, sandboxed execution, workflow checkpointing

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Random Forest, neural networks, recommendation systems, anomaly detection, SQL

CERTIFICATIONS

Andrew Ng Machine Learning Specialization: Supervised Machine Learning; Advanced Learning Algorithms; Unsupervised Learning, Recommenders, and Reinforcement Learning - DeepLearning.AI / Coursera

Data Analysis with Python; Data Science Methodology - IBM / Coursera

LINKS

AI SWE Agent: github.com/purbiyarakesh33/ai-swe-agent

Amazon Recommender: [Live demo](#) | [GitHub](#)

Fraud Detection: [Live demo](#) | [GitHub](#)

Customer Churn: [Live demo](#) | [GitHub](#)